

TAICHUNG SHUINAN AP, Taiwan

WMO: 467510

Lat: 24.186N

Lon: 120.654E

Elev: 377

StdP: 14.50

Time Zone: 8.00 (E08)

Period: 86-99

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.3	48.6	40.8	38.0	48.7	43.1	41.6	50.7	22.5	61.3	20.5	60.6	4.1	10	0.366

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.3	93.6	82.1	92.9	82.0	91.6	81.5	84.1	91.8	83.2	90.9	82.3	90.0	9.8	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.2	170.1	90.3	80.9	162.6	89.1	80.3	159.5	88.5	48.7	91.6	47.4	91.0	46.6	90.2	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.7	17.9	16.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
			DB												
			WB	38.3	86.2	4.1	0.9	35.3	86.9	32.9	87.4	30.6	87.9	27.6	88.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	74.4	62.5	63.1	66.9	73.9	78.7	82.5	84.5	84.2	81.6	77.3	71.5	65.1	(6)
(7)		DBStd	9.08	4.65	5.52	6.36	5.81	4.01	3.14	2.20	2.46	3.11	3.50	4.66	4.56	(7)
(8)		HDD50	1	0	0	1	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	326	104	93	56	9	0	0	0	0	0	0	8	56	(9)
(10)		CDD50	8894	387	367	525	715	891	976	1068	1060	948	846	644	467	(10)
(11)		CDD65	3744	26	39	115	275	426	526	603	595	498	381	202	58	(11)
(12)		CDH74	35475	146	232	638	1921	3644	5707	7149	6944	4974	2772	1057	291	(12)
(13)		CDH80	13730	13	35	147	567	1305	2345	3151	2982	1952	948	260	25	(13)
(14)	Wind	WSAvg	6.6	8.0	7.5	7.2	6.0	5.2	5.7	5.1	5.6	5.9	7.5	7.7	7.6	(14)
(15)	Precipitation	PrecAvg	63.1	1.3	3.0	3.7	4.6	8.4	13.7	9.2	12.4	4.3	0.6	0.7	0.8	(15)
(16)		PrecMax	92.0	4.8	15.1	13.9	18.6	21.3	35.0	25.3	30.9	22.7	5.1	2.9	3.3	(16)
(17)		PrecMin	38.5	0.2	0.1	0.1	0.0	0.2	1.8	0.8	2.9	0.2	0.0	0.0	0.0	(17)
(18)		PrecStd	13.9	1.1	3.7	2.8	3.9	4.7	8.6	5.9	6.7	4.3	0.9	0.8	0.9	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.6	83.9	86.6	90.0	93.0	95.0	95.4	93.5	91.3	88.2	82.8	(19)	
(20)		MCWB	70.4	72.7	74.9	79.0	80.9	82.0	82.7	83.0	82.2	79.8	77.5	72.6	(20)	
(21)		2%	DB	78.0	80.5	84.0	88.0	90.3	93.1	93.6	93.5	91.8	89.3	85.8	80.2	(21)
(22)		MCWB	68.0	69.9	74.1	78.1	79.9	81.8	82.2	82.3	81.8	79.0	75.4	70.0	(22)	
(23)		5%	DB	74.9	76.8	80.9	85.9	88.8	91.5	92.7	92.1	90.7	87.6	82.8	77.3	(23)
(24)		MCWB	66.1	68.2	72.3	76.9	79.4	81.1	82.0	81.9	81.2	77.9	73.8	68.1	(24)	
(25)		10%	DB	71.6	73.3	77.4	83.7	87.1	89.8	91.4	91.1	89.2	85.6	80.4	75.0	(25)
(26)		MCWB	64.2	66.4	70.3	75.7	78.8	80.5	81.6	81.5	80.2	76.5	72.0	66.7	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	72.4	74.4	77.0	80.7	82.7	84.5	85.5	85.2	84.1	81.6	79.3	73.4	(27)
(28)		MCDB	80.2	81.8	84.9	88.0	90.0	92.1	92.5	92.7	91.0	88.9	86.3	81.7	(28)	
(29)		2%	WB	68.7	71.4	74.8	78.9	81.0	83.0	84.2	83.7	82.8	79.8	76.4	71.1	(29)
(30)		MCDB	75.9	77.8	82.3	86.3	88.4	90.9	91.8	91.8	90.2	87.2	83.8	78.3	(30)	
(31)		5%	WB	66.8	69.0	73.0	77.4	79.9	82.0	83.1	82.7	81.9	78.6	74.5	69.2	(31)
(32)		MCDB	73.6	75.3	79.8	84.4	86.9	89.7	90.8	90.7	89.2	85.7	81.5	76.0	(32)	
(33)		10%	WB	64.9	66.7	71.0	75.9	78.9	80.9	81.9	81.8	80.7	77.4	72.7	67.2	(33)
(34)		MCDB	71.1	72.7	76.5	82.2	85.4	88.4	89.6	89.4	87.6	84.2	78.9	73.2	(34)	
(35)	Mean Daily Temperature Range	MDBR	18.6	18.2	16.8	15.7	14.7	14.1	14.3	13.9	15.1	17.2	17.9	19.5	(35)	
(36)		5% DB	MCDBR	23.1	23.5	21.5	17.9	16.8	15.9	15.9	15.6	16.2	18.6	20.0	22.6	(36)
(37)		MCWBR	14.7	14.4	13.1	10.5	9.2	8.4	8.3	7.8	8.9	10.8	12.0	14.1	(37)	
(38)		5% WB	MCDBR	21.3	21.8	20.7	17.1	15.9	15.2	15.4	14.9	15.4	16.7	19.2	20.9	(38)
(39)		MCWBR	14.3	14.2	13.3	10.7	9.6	8.7	8.7	8.2	9.0	10.4	12.1	14.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.491	0.546	0.666	0.677	0.580	0.533	0.546	0.580	0.572	0.535	0.501	0.492	(40)	
(41)		taud	2.009	1.893	1.652	1.682	1.945	2.094	2.047	1.943	1.939	1.986	2.065	2.024	(41)	
(42)		Ebn at Noon	239	235	213	215	237	246	243	235	233	235	234	232	(42)	
(43)		Edh at Noon	50	60	80	79	61	52	54	60	59	54	47	48	(43)	
(44)	All-Sky Solar Radiation	RadAvg	920	1018	1172	1287	1476	1627	1852	1677	1543	1364	1025	910	(44)	
(45)		RadStd	126	148	131	174	222	178	116	163	146	115	97	110	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (28 neighbors)	+0.45	N/A	N/A	+0.40	N/A	N/A	N/A	N/A	+229	+202

Nomenclature: See separate page